SCIENCE & INNOVATION FOR SUSTAINABILITY

PROJECT RESEARCH LOGBOOK & DAILY WORK DIARY

Official Research Record Template for Handwritten Project Notebook

Project Name:	KRISHI-DRISHTI (Krishi-Drishti)
Working Title:	An Offline-First Edge-AI Plant Health Screening and Microclimate Monitoring System for Sustainable Agriculture
Main Research Theme:	Science and Innovation for Sustainability
Sub-Theme:	Food, Agriculture and Health
Lead Researcher (Student 1):	_____
Co-Worker (Student 2):	_____
Class & Section:	Class 9 [_____]
School Name & Place:	_____
District & State:	Vaishali, Bihar — PIN: [_____]
Guide Teacher / Mentor:	_____
Diary Started On:	____ / ____ / 2026
Diary Concluded On:	____ / ____ / 2026

Important Note for Student Researchers: *A science logbook is like a daily diary of your project. Write in it on the exact day you do any work. If a circuit wire gets hot, if a sensor gives zero, or if code crashes, write it down honestly! Judges and reviewers look for honest learning, troubleshooting, and real student work—never copy or make up fake entries at the end.*

HOW TO MAINTAIN THIS PROJECT LOGBOOK

Our guide teacher gave us four golden rules when we began writing our research notebook:

1. **Write As You Go:** Never wait until the weekend or next month to fill this diary. Keep it open on your workbench beside your soldering iron or breadboard.
2. **Record Actual Numbers:** Do not just say 'tested sensor today'. Write: 'DHT11 gave 22.4°C and 88% humidity at 7:00 AM in morning fog'.
3. **Mistakes Are Welcome:** Every time we burnt a wire, got a brownout crash, or got an 87% false alarm on a tea leaf, we wrote down what happened and how we fixed it.
4. **Teacher Signatures:** Show this notebook to your science teacher every Friday so they can check your progress and sign the bottom.

SECTION A: INITIAL PROJECT PLANNING & RESEARCH DESIGN

Initial Problem Observed & Investigated:
(Describe what sparked your project — e.g., tracking viral agricultural news of severe tomato blight in Kolar/Nashik and potato late blight across Samastipur, Vaishali, and Agra; seeing farmers spray blind chemicals after leaves were already black)
Our Central Research Question:
(Write the main question you want to answer with your experiment)
Our Working Hypothesis:
(What do you predict will happen when using an 8 cm shroud and humidity sensor?)
Project Scope & Limits:
Target crops: Potato & Tomato | Target diseases: Early Blight, Late Blight, Healthy | Hardware cost: Rs 1,480 (Under Rs 1,500) | 100% Offline.

SECTION B: MATERIALS, SENSORS & EQUIPMENT REGISTER

We kept track of every single rupee we spent from our pocket money and school science kit:

S.No.	Component / Tool	Key Specification	Where We Got It	Date	Cost (Rs)
1	ESP32 DevKit V1	Dual Core 240MHz, 30-pin board	Electronics shop, Patna	12/07/2026	Rs 420
2	AI-Thinker ESP32-CAM	ESP32-S + OV5640 5MP sensor	Online hobby parts store	15/07/2026	Rs 540
3	SSD1306 OLED Display	0.96 inch, 128x64 pixels, I2C	Electronics shop, Patna	18/07/2026	Rs 190
4	DHT11 Sensor	Temp (0–50°C), RH (20–90%)	School physics lab kit	20/07/2026	Rs 95
5	Pushbuttons & Resistors	2 tact buttons, 10k resistor	Lab junk box	22/07/2026	Rs 20
6	MicroSD Card (16GB)	Class 10 high-speed FAT32	Local mobile repair shop	25/07/2026	Rs 160
7	Optical Shroud Tube	8 cm black 3D cone / PVC pipe	DIY made in school workshop	05/08/2026	Rs 45
8	Electrolytic Capacitor	470 uF, 16V (fixes brownouts)	Electronics spare board	10/08/2026	Rs 10
9	Samsung Galaxy A03	Android 13, 3GB RAM (Pocket AI)	Family phone (borrowed)	Available	Rs 0 (Existing)
10	USB Power Bank	5,000 mAh portable battery	Borrowed from home	Available	Rs 0 (Existing)

SECTION C: OUR FIVE-MONTH WORK TIMELINE

Month 1 (July 2026): *Investigated multi-state blight reports from agricultural news and social media; surveyed local vegetable growers; bought ESP32 and breadboards.*
Month 2 (August 2026): *Wired OLED display and DHT11; fixed brownout crashes on ESP32-CAM using 470uF capacitor.*
Month 3 (September 2026): *Built 8 cm optical shroud; set up offline Wi-Fi hotspot; tested raw photos.*
Month 4 (October 2026): *Cleaned training dataset with SHA-256 hashes; trained MobileNetV3 TFLite (2.03 MB); discovered tea leaf error.*
Month 5 (November 2026): *Evaluated 60 botanical leaves and 93 PlantDoc photos; compiled charts and final report.*

SECTION D: DAILY RESEARCH DIARY SHEET — ENTRY LOG #1

Real-time handwritten laboratory and field log record.

Date: 12 / 08 / 2026	**Time: 03:30 PM**	**Duration: 2.5 Hours**
Activity Performed: Testing I2C wiring between ESP32 DevKit V1 and SSD1306 0.96 inch OLED screen on breadboard.		
Observations Made: At first, the screen stayed completely blank and dark. The serial monitor printed 'I2C Device Not Found at 0x3C'.		
Measurements / Numerical Data: VCC = 3.32V, GND = 0V, Bus Frequency = 400 kHz Fast Mode. GPIO 21 (SDA) and GPIO 22 (SCL).		
Problem / Glitch Encountered: We accidentally crossed the jumper wires: SDA was plugged into GPIO 22 and SCL into GPIO 21.		
Troubleshooting & Action Taken: Swapped the wires back to GPIO 21 (SDA) and GPIO 22 (SCL). Reloaded the sketch and the blue pixels immediately lit up!		
What I Learned Scientifically Today: I2C is an address-based bus. If clock and data are swapped, the master controller cannot receive ACK signals from the slave.		
Student Signature: _____		**Guide Teacher Signature:** _____

SECTION D: DAILY RESEARCH DIARY SHEET — ENTRY LOG #2

Real-time handwritten laboratory and field log record.

Date: 04 / 09 / 2026	**Time: 11:00 AM**	**Duration: 3.0 Hours**
Activity Performed: Testing camera flash and image capture outdoors in bright noon sunlight in school garden.		
Observations Made: When holding the camera by hand, sunlight caused terrible glare. Also, every time the flash LED fired, the ESP32-CAM rebooted!		
Measurements / Numerical Data: Sunlight lux > 85,000 lux. Power rail dipped from 3.3V down to 2.4V for 15 milliseconds during flash + Wi-Fi burst.		
Problem / Glitch Encountered: The board suffered from brownout reset because Wi-Fi transmission and LED flash draw over 300 mA simultaneously.		
Troubleshooting & Action Taken: Soldered a 470 uF electrolytic capacitor directly across the 5V and GND pins. Added an 8 cm dark tube (shroud) to block sun glare.		
What I Learned Scientifically Today: Transient current spikes cause voltage drops on cheap power supplies. Decoupling capacitors act like tiny water tanks to smooth power.		
Student Signature: _____		**Guide Teacher Signature:** _____

SECTION D: DAILY RESEARCH DIARY SHEET — ENTRY LOG #3

Real-time handwritten laboratory and field log record.

Date: 18 / 10 / 2026	**Time: 04:15 PM**	**Duration: 4.0 Hours**
Activity Performed: Testing the trained MobileNetV3 Float16 model on 60 fresh botanical leaves and 1 unseen alien tea leaf.		
Observations Made: The model got 53 out of 60 botanical leaves correct. But on the tea leaf, it predicted 'Potato Late Blight' with 87% confidence!		
Measurements / Numerical Data: Botanical strict accuracy = 53/60 (88.33%). Pathogen match = 54/60 (90.0%). Tea leaf confidence = 87.0% (False Acceptance).		
Problem / Glitch Encountered: Closed-set neural networks have no 'I don't know' class. They force any visual input into one of the six trained classes.		
Troubleshooting & Action Taken: Added Shannon Entropy gating (H > 1.45 nats) in our Python daemon to flag high-entropy distributions as UNCERTAIN.		
What I Learned Scientifically Today: An optical shroud standardizes focal distance, but it cannot make a simple AI model botanically omniscient. Testing out-of-distribution cases is essential.		
Student Signature: _____		**Guide Teacher Signature:** _____

SECTION E: EXPERIMENTAL OBSERVATION LOG SHEETS

Experiment 1: Why We Needed an Optical Shroud (Focal Distance Test)

We tested what happens when holding the camera by hand at different distances vs attaching our 8 cm shroud:

Trial	Distance	Shroud?	Resolution	What the Camera Saw	AI Diagnosis Result
1	3 cm	No	1600x1200	Blurry green smudge; cannot see leaf veins	UNCERTAIN (Entropy > 1.45)
2	5 cm	No	1600x1200	Slightly out of focus; edges hazy	Weak guess (52% confidence)
3	8 cm	Yes (Shroud)	1600x1200	Razor sharp; concentric rings clearly visible	Tomato Early Blight (99.8%)
4	10 cm	Yes (Shroud)	1600x1200	Even lighting; dark water-rot visible	Potato Late Blight (100.0%)
5	15 cm	No	1600x1200	Leaf is tiny; dry mud fills 70% of photo	REJECTED (Foliar ratio < 12%)
6	20 cm	No	1600x1200	Student's hand and slippers in photo	NON_LEAF (Filter tripped)

Experiment 2: Checking Temperature & Humidity Inside Leaf Foliage

We took DHT11 readings in local potato fields at different times of day to check Late Blight risk:

Reading	Time & Weather Condition	Temp	RH (%)	Blight Risk Level	Prescription Advice
R-1	07:00 AM (Cold, dripping morning fog)	18.5°C	92.0%	SEVERE (Wallin envelope met)	High risk! Check lower leaves for rot
R-2	10:30 AM (Sun coming out; leaves damp)	22.1°C	86.5%	WARNING (Favorable for blight)	Do not use sprinkler irrigation
R-3	01:30 PM (Bright, hot, dry afternoon)	27.4°C	62.0%	SAFE (Too dry for spore growth)	Do not spray fungicides; air is dry
R-4	05:30 PM (Sun setting; evening dew)	20.2°C	88.0%	WARNING (Dew forming)	Apply Trichoderma bio-spray if needed

SECTION F: OUR COMPLETE TROUBLESHOOTING & FAILURE LOG

We kept this record of all the things that broke during our project and how we solved them:

Date	What Broke / Failed	Why It Happened (Root Cause)	How We Fixed It
12/08	**OLED screen stayed dark**	SDA and SCL jumper wires were reversed	Rewired SDA to GPIO 21, SCL to GPIO 22
24/08	**Buttons clicked repeatedly**	Mechanical switch bounce sent 5 pulses per click	Added 50ms software debounce timer in code
04/09	**Camera board rebooted on flash**	Wi-Fi + LED pulled 3.3V rail down to 2.4V	Soldered 470 uF capacitor across 5V and GND
19/09	**Photos were dark at the corners**	Shroud tube was too narrow; blocked camera lens angle	Widened inner diameter from 2.5 cm to 4.0 cm
08/10	**Phone couldn't find camera Wi-Fi**	ESP32-CAM was set to 5GHz channel by mistake	Configured standard 802.11 b/g/n 2.4GHz channel 1
18/10	**Tea leaf predicted as blight 87%**	Closed-set model forced alien leaf into 1 of 6 classes	Added Shannon entropy filter (H > 1.45 nats)
26/10	**Healthy leaf called Early Blight**	Angled torch cast a harsh dark shadow across leaf vein	Added acrylic diffusion disc over LED light

SECTION G: GUIDE TEACHER'S PERIODIC ASSESSMENT

Evaluation Parameter	Max Marks	Marks Awarded	Teacher's Remarks
1. Originality of problem selection & real-world relevance	10	____ / 10	Genuinely addresses severe agricultural blight issues
2. Scientific methodology & experimental rigor	10	____ / 10	Systematic focal tests, microclimate checks & gating
3. Hardware prototyping & practical engineering	10	____ / 10	Low-cost dual-ESP32 build, brownout resolved
4. Honesty in recording errors and failure analysis	10	____ / 10	Recorded tea leaf OOD trap & drop-shadow honest failures
5. Regularity of logbook maintenance	10	____ / 10	Entries recorded consistently across all project phases
Total Project Score	**50**	**____ / 50**	**Verified & Recommended for Project Submission**

Guide Teacher Final Comment:

_____

Teacher Signature: _____ **Date:** ____ / ____ / 2026 **Institution Seal:** [STAMP]